

Unit 3

Physical Layer and its Design Issues

(7 Hours)

3.1 Introduction, Design Issues and Duties of Physical Layer

3.2 Transmission Media: Guided (Twisted Pair, Coaxial Cable, Optical Fiber), Unguided (Radio Wave, Microwave, Infrared and Satellite)

3.3 Ethernet Cable Standards

3.4 Ethernet, Fast-Ethernet and GigaEthernet

3.5 IEEE 802.11 (Wi-Fi) and IEEE 802.15.1 (Bluetooth) Standards

3.6 Signals

3.7 Transmission Impairment: Attenuation, Distortion and Noise

3.8 Multiplexing: TDM, FDM and WDM

3.9 Switching: Circuit Switching, Message Switching, Packet Switching, Virtual Circuit Switching

3.10 Network Performance: Bandwidth, Throughput, Latency, Bandwidth-Delay Product, Jitter

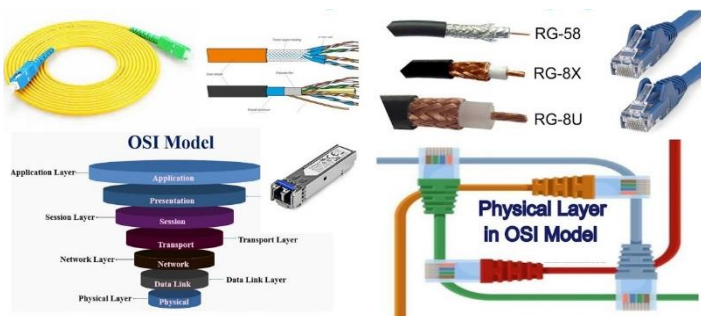
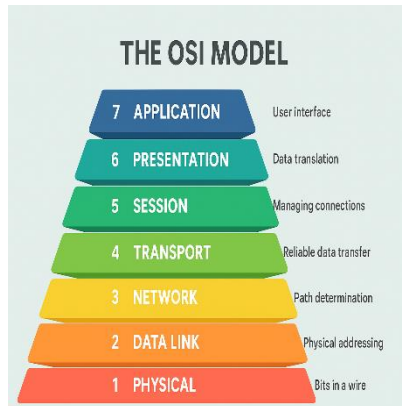
3.1 Introduction, Design Issues and Duties of Physical Layer

The **Physical Layer** is the **first and lowest layer** of the OSI model.

It is responsible for the **actual transmission of raw bits** from one device to another through a physical medium.

This layer deals with:

- cables
- connectors
- voltage levels
- signals
- data rate
- transmission mode
- physical network devices



Simple meaning

If upper layers prepare the data, the physical layer actually **puts the bits on the wire or air**.

Duties of Physical Layer

The main duties of the physical layer are:

1. Transmission of Raw Bits

It sends **0s and 1s** over the communication channel.

2. Representation of Bits

It defines how bits are represented:

- electrical signals
- light pulses
- radio waves

3. Data Rate Control

It specifies how many bits are transmitted per second.

4. Bit Synchronization

It ensures sender and receiver remain synchronized.

5. Line Configuration

It decides whether communication is:

- point-to-point
- multipoint

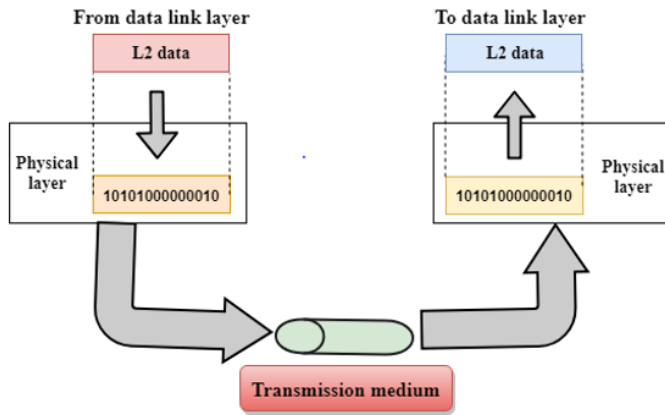
6. Physical Topology

It supports physical arrangement such as:

- bus
- star
- ring

It defines:

- simplex
- half-duplex
- full-duplex



Design Issues of Physical Layer

Important design issues are:

- how bits should be represented
- how sender and receiver synchronize
- what transmission medium should be used
- how much data rate is supported
- how connectors and interfaces are designed
- how signals travel over long distance
- how to reduce interference and loss

3.2 Transmission Media

Introduction

Transmission media means the path or channel through which data travels from the sender to the receiver.

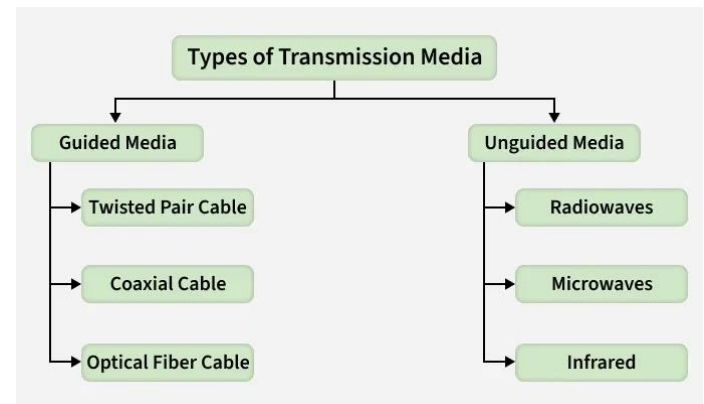


In a communication system, data cannot move by itself. It needs **some medium to carry signals**.

Example:

- when two computers are connected by cable, the **cable is the transmission medium**
- when mobile phones communicate wirelessly, air acts as the transmission medium

Transmission media are mainly divided into **two types**:



1. Guided Media

2. Unguided Media

1. Guided Media

In **guided media**, signals travel through a **physical path**.

The signal is guided inside a cable or wire.

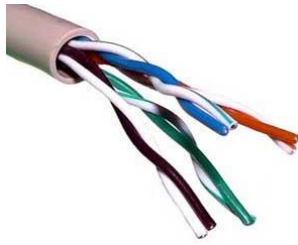
Types of Guided Media:

- Twisted Pair Cable
- Coaxial Cable
- Optical Fiber

1.1 Twisted Pair Cable

A **twisted pair cable** consists of **two insulated copper wires twisted together**.

The twisting helps reduce noise and interference.

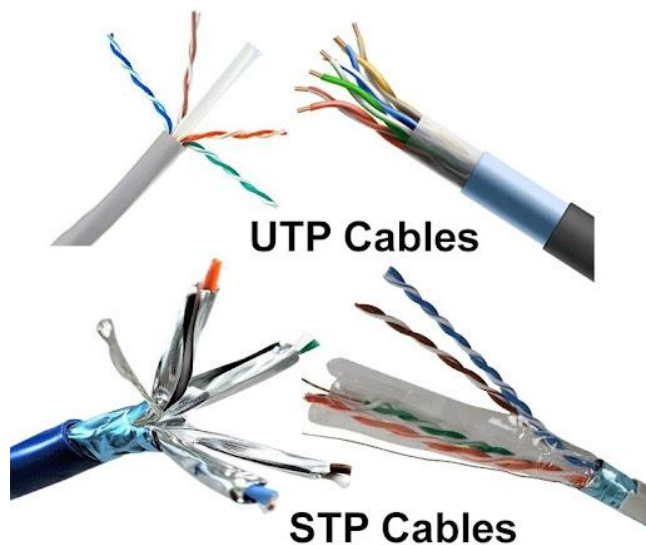


Types:

- UTP (Unshielded Twisted Pair)
- STP (Shielded Twisted Pair)

Uses:

- telephone lines
- LAN networks
- Ethernet connections
- office networking



Advantages:

- low cost

- easy to install
- flexible
- widely used

Disadvantages:

- limited distance
- more affected by noise than fiber
- lower bandwidth than optical fiber

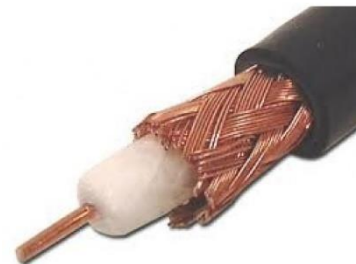
Example:

In a college computer lab, PCs connected to a switch using LAN cable usually use twisted pair cable.

1.2 Coaxial Cable

A **coaxial cable** has:

- one central copper conductor
- insulation layer
- metallic shield
- outer protective cover



It provides better shielding than twisted pair cable.

Uses:

- cable television
- CCTV
- broadband internet
- radio communication



Advantages:

- better protection from interference
- higher bandwidth than twisted pair

- stronger and durable

Disadvantages:

- more expensive than twisted pair
- thicker and less flexible
- installation is harder

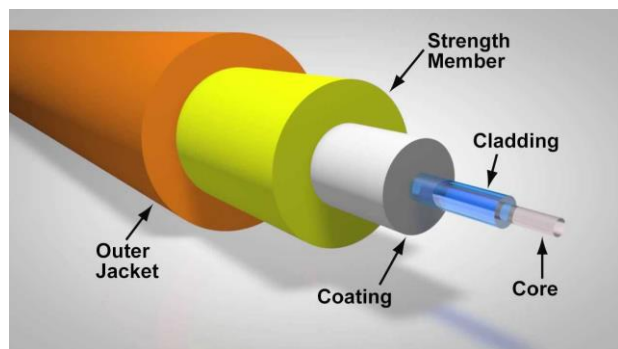
Example:

Cable TV line entering a home is commonly a coaxial cable.

1.3 Optical Fiber

An **optical fiber** is a thin strand of glass or plastic that carries data in the form of **light signals**.

It is the fastest and most secure guided medium.



Types:

- **Single-mode fiber**
- **Multi-mode fiber**

Uses:

- high-speed internet
- backbone networks
- long-distance communication
- telephone networks



Advantages:

- very high bandwidth

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- very high speed

- low attenuation
- not affected by electromagnetic interference
- more secure

Disadvantages:

- expensive
- difficult to install
- fragile compared to copper wire



Example:

Internet service providers use optical fiber to connect cities and major networks.

2. Unguided Media

In **unguided media**, signals travel through **air or free space**.

No physical cable is required.

These are also called **wireless transmission media**.

Types of Unguided Media:

- Radio Wave
- Microwave
- Infrared
- Satellite

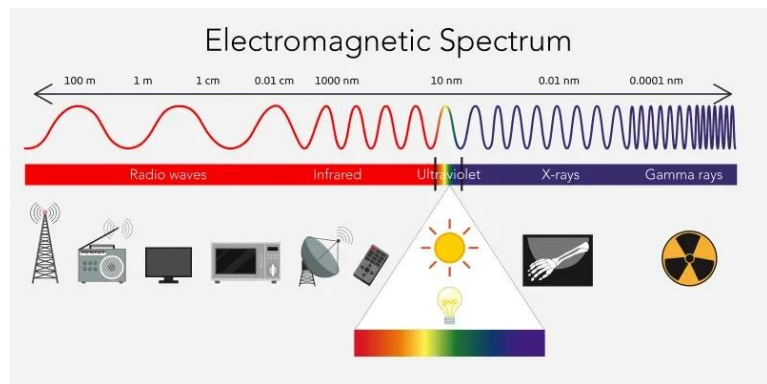
Guided Vs Unguided Transmission Media



2.1 Radio Wave

Radio waves are electromagnetic waves used for wireless communication.

They can travel through air and can pass through walls.



Uses:

- radio broadcasting
- television broadcasting
- Wi-Fi
- mobile communication



Advantages:

- no wires required
- supports mobility
- can cover large areas

Disadvantages:

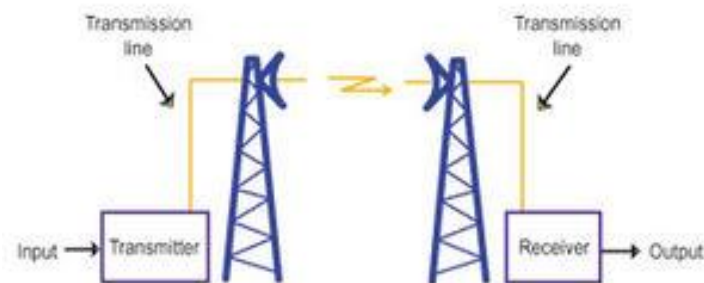
- interference possible
- lower security
- quality may be affected by environment

Example:

Wi-Fi in a home uses radio waves.

2.2 Microwave

Microwaves are high-frequency radio waves used for long-distance communication.



They usually require **line-of-sight transmission**, meaning there must be a clear path between sender and receiver.

Uses:

- mobile towers
- satellite communication
- long-distance telephone links
- point-to-point communication

Advantages:

- high bandwidth
- good for long-distance communication
- supports large data transfer

Disadvantages:

- affected by weather
- needs line of sight
- installation cost may be high

Example:

Communication between two cell towers often uses microwave links.

2.3 Infrared

Infrared is a short-range wireless medium that uses infrared light waves.

It cannot pass through walls.

Uses:

- TV remote control
- short-range device communication
- some wireless peripherals



Advantages:

- simple
- low cost
- secure for room-level communication

Disadvantages:

- very short range
- cannot pass through obstacles
- not suitable for long distance

Example:

A television remote uses infrared signals.

2.4 Satellite

In **satellite communication**, signals are sent from Earth to a satellite in space and then back to another location on Earth.

It generally uses microwaves.

Uses:

- TV broadcasting
- GPS
- weather monitoring
- internet in remote areas
- military communication



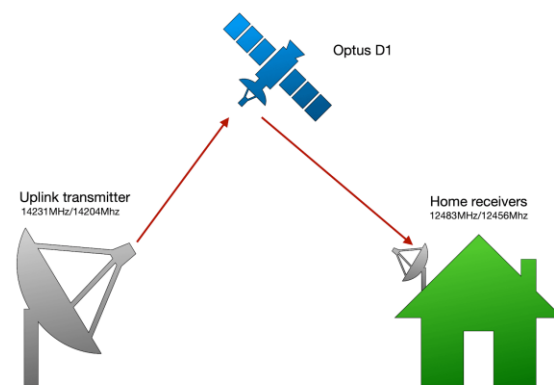
Advantages:

- very wide coverage
- useful in remote and mountainous areas
- supports long-distance communication



Disadvantages:

- expensive
- high latency
- affected by weather in some cases



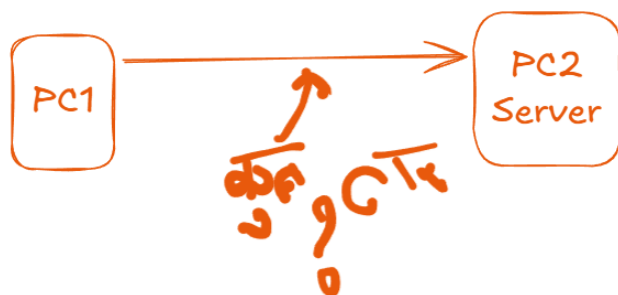
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Example:

DTH television and global GPS systems use satellites.

1. Introduction

Ethernet cable standards define the proper arrangement of wires inside an Ethernet cable so that devices can communicate correctly in a network.

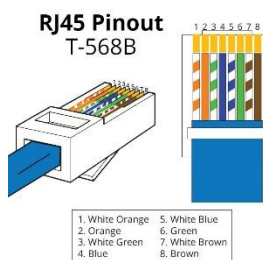


Ethernet cables are commonly used in:

- LAN
- office networking
- school labs
- home routers and switches

The most common Ethernet cable uses an **RJ45 connector** with **8 wires**.

To make the cable correctly, the wires must follow standard color codes.



The two main standards are:

- **T568A**
- **T568B**

1. T568A Standard

In the **T568A** standard, the 8 wires are arranged in this order:

1. **White/Green**

2. **Green**

3. **White/Orange**

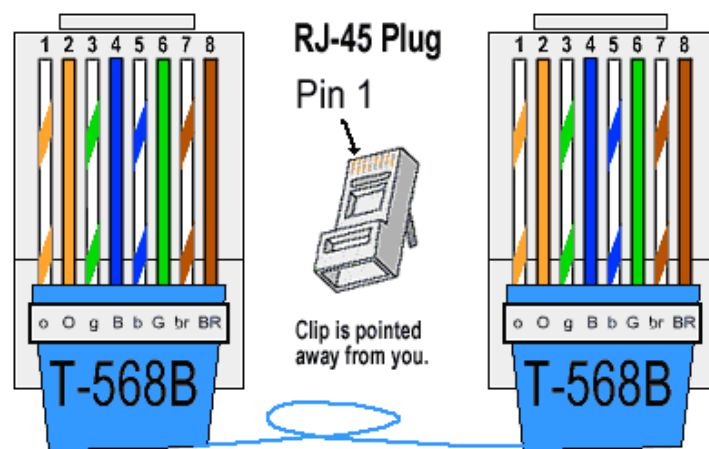
4. **Blue**

5. **White/Blue**

6. **Orange**

7. **White/Brown**

8. **Brown**



2. T568B Standard

In the **T568B** standard, the 8 wires are arranged in this order:

1. **White/Orange**

2. **Orange**

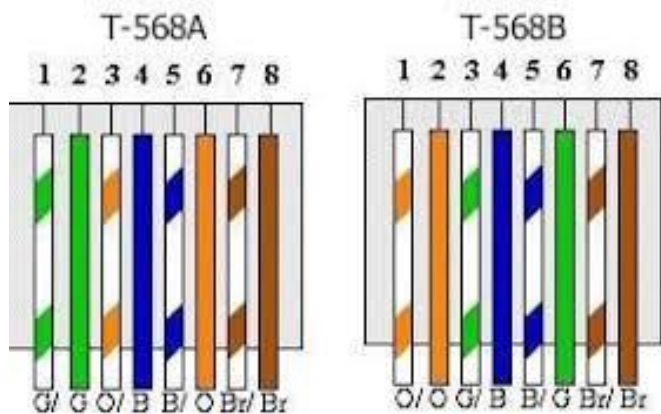
3. **White/Green**

4. **Blue**

5. **White/Blue**

6. **Green**

7. **White/Brown**



3. Why These Standards are Important

Ethernet cable standards are important because they:

- **ensure** correct communication
- reduce wiring **mistakes**
- provide **compatibility**
- help in making **standard** network cables
- make troubleshooting easier

If the wire arrangement is wrong, the cable may not work properly.

4. Types of Ethernet Cables Based on Standards

A. Straight-Through Cable

A **straight-through cable** has the **same standard on both ends**.

Examples:

- T568A to T568A
- T568B to T568B

Uses:

- PC to switch

- PC to hub

- router to switch

Meaning:

The wire order is the same at both ends.

B. Cross-Over Cable

A **cross-over cable** has **different standards on each end**.

Example:

- one end = T568A
- other end = T568B

Uses:

- PC to PC
- switch to switch
- hub to hub
(mainly in older networks)

Meaning:

Transmit and receive pairs are crossed.

5. Ethernet Cable Categories

Ethernet cables also come in different **categories**, which define speed and performance.

Common categories:

- Cat5
- **Cat5e**
- **Cat6**
- Cat6a
- Cat7

- higher category = better speed and performance

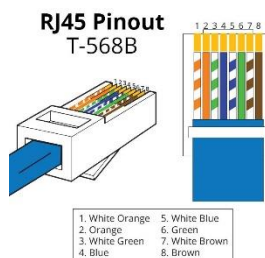
Example:

- **Cat5e** supports up to 1 Gbps
- **Cat6** supports higher speed and better performance

6. RJ45 Connector

The **RJ45 connector** is the connector used at the end of Ethernet cable.

It has **8 pins**, and each pin connects to one wire of the twisted pair cable.



Use:

It is plugged into:

- NIC

3.4 Ethernet, Fast-Ethernet and GigaEthernet

Ethernet is one of the most widely used technologies in computer networks, especially in **LAN (Local Area Network)**.

It is used to connect:

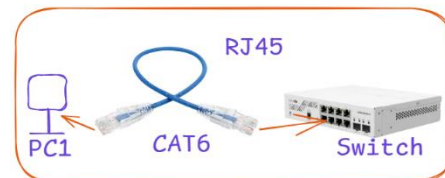
- computers
- switches
- routers
- printers
- servers

- switch
- router
- hub

7. Simple Example

Suppose a student wants to connect a computer to a switch in a lab.

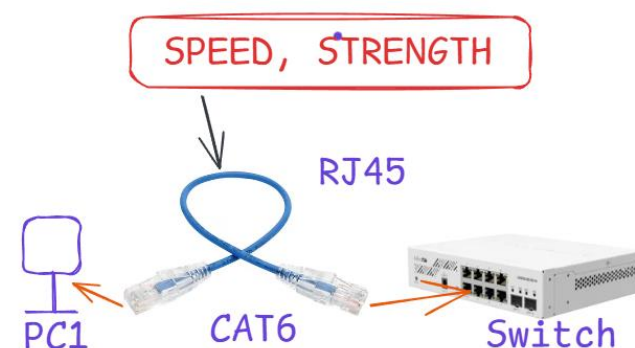
He should usually use a **straight-through cable**, such as:



- T568B on one end
- T568B on the other end

If he wants to connect one old switch directly to another old switch, he may use a **cross-over cable**:

- T568A on one end
- T568B on the other end



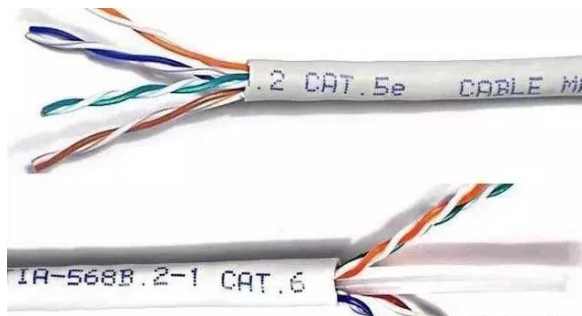
Ethernet defines how data is transmitted in a wired network.

As network demand increased, faster versions of Ethernet were developed:

- **Ethernet**
- **Fast Ethernet**

• Gigabit Ethernet

The main difference among them is **speed**.



1. Ethernet

The original **Ethernet** provides a speed of:

10 Mbps

(10 megabits per second)

It is the basic form of Ethernet used in early LANs.

Features:

- speed = 10 Mbps
- used in small or older networks
- suitable for basic data transfer

Example:

Older office computers connected in a LAN with 10 Mbps speed.

Limitation:

As internet use, file transfer, and multimedia increased, 10 Mbps became too slow.

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2. Fast Ethernet

Fast Ethernet is an improved version of Ethernet.

It provides a speed of:

100 Mbps

(100 megabits per second)

This is **10 times faster** than traditional Ethernet.

Features:

- speed = 100 Mbps
- better for office networks
- supports larger data transfer
- widely used in many LANs

Example:

A school computer lab where many computers share files and internet.

Advantage over Ethernet:

It reduces transfer time and improves performance.

3. Gigabit Ethernet

Gigabit Ethernet is a much faster version of Ethernet.

It provides a speed of:

1000 Mbps = 1 Gbps

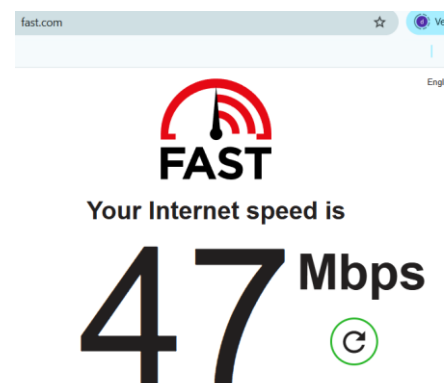
(1 gigabit per second)

This is:

- **100 times faster than Ethernet**
- **10 times faster than Fast Ethernet**

Features:

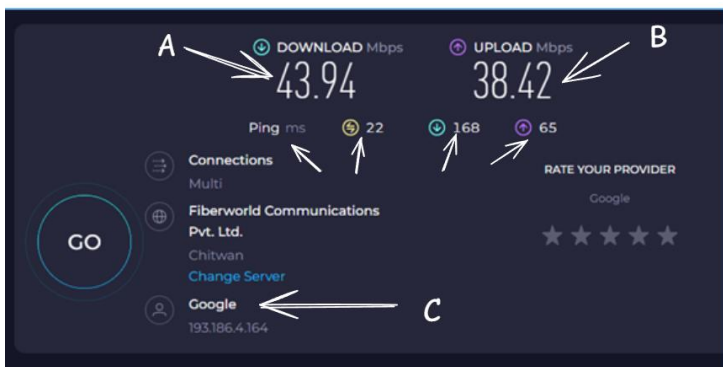
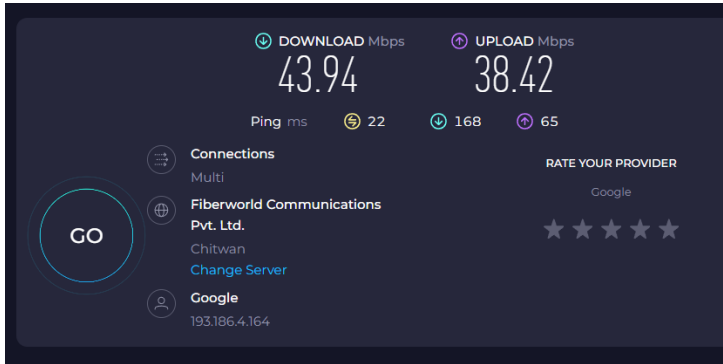
- speed = 1 Gbps



- used in modern LANs
- suitable for high-speed communication
- supports video, server connection, large file transfer

Example:

Modern colleges, data centers, offices, and backbone LAN connections often use Gigabit Ethernet.



4. Simple Speed Comparison

Type	Speed
Ethernet	10 Mbps
Fast Ethernet	100 Mbps
Gigabit Ethernet	1000 Mbps or 1 Gbps

5. Easy Analogy

Imagine three roads for vehicles:

- **Ethernet** = narrow road
- **Fast Ethernet** = wider road
- **Gigabit Ethernet** = highway

The wider the road, the more vehicles can pass quickly.

In the same way:

- higher Ethernet speed means more data can travel in less time

3.5 IEEE 802.11 (Wi-Fi) and IEEE 802.15.1(Bluetooth) Standards

Introduction

IEEE is an international organization that develops standards for networking and communication technologies.



In computer networks, different wireless technologies follow different IEEE standards.

Two important wireless standards are:

- **IEEE 802.11 for Wi-Fi**
- **IEEE 802.15.1 for Bluetooth**

These standards define how wireless devices communicate with each other.

IEEE 802.11 is the standard used for **wireless local area network (WLAN)**, commonly known as **Wi-Fi**.

Wi-Fi allows devices to connect to a network or the internet **without using cables**.

Common devices using Wi-Fi:

- laptop
- smartphone
- tablet
- smart TV
- printer

Main features of Wi-Fi:

- wireless communication
- medium to high range
- high data transfer speed
- used for internet access
- used in homes, offices, schools, colleges, and public places

Uses of Wi-Fi:

- connecting mobile and laptop to internet
- wireless office networking
- smart home devices
- wireless campus network

Example:

When a student connects a laptop to the college router without cable, that is Wi-Fi communication under **IEEE 802.11**.

IEEE 802.15.1 is the standard used for **Bluetooth** communication.

Bluetooth is used for **short-range wireless personal communication**.

It is mainly used to connect nearby devices directly.

Common devices using Bluetooth:

- mobile phone
- wireless headphones
- speaker
- keyboard
- mouse
- smartwatch

Main features of Bluetooth:

- short-range communication
- low power consumption
- used for personal area network (PAN)
- device-to-device connection
- does not need internet for direct pairing

Uses of Bluetooth:

- mobile to headphone connection
- file sharing between two phones
- wireless mouse and keyboard connection
- smartwatch pairing

Example:

When a person connects a Bluetooth earbud to a mobile phone, that communication follows **IEEE 802.15.1** standard.

TYPICAL SYSTEM PARAMETERS OF THE WIRELESS PROTOCOLS				
Standard	Bluetooth	UWB	ZigBee	Wi-Fi
IEEE Spec.	802.15.1	802.15.3	802.15.4	802.11a/b/g
Max data rate (Mbit/s)	0.72	110*	0.25	54
Bit time (μ s)	1.39	0.009	4	0.0185
Max data payload (bytes)	339 (DH5)	2044	102	2312
Max overhead (bytes)	158/8	42	31	58
Coding efficiency ⁺ (%)	94.41	97.94	76.52	97.18
* Unapproved 802.15.3a.		+ Where the data is 10K bytes.		

3. Difference Between Wi-Fi and Bluetooth

Basis	Wi-Fi (IEEE 802.11)	Bluetooth (IEEE 802.15.1)
Full form	Wireless LAN	Wireless Personal Area Network
Range	Larger	Shorter
Speed	Higher	Lower
Power consumption	More	Less
Main use	Internet and network access	Device-to-device connection
Example	Laptop connected to router	Phone connected to headset

4. Simple Analogy

Think of:

- **Wi-Fi** like a **school bus** that carries many students over a longer distance

This means:

- Wi-Fi connects devices over a wider area and carries more data
- Bluetooth is for short-distance personal use

5. Simple Examples

Example of Wi-Fi

In a college:

- one wireless router is installed
- students connect laptops and phones
- all devices access internet without cable

This is **IEEE 802.11 Wi-Fi**

Example of Bluetooth

A teacher connects:

- phone to wireless speaker
- laptop to Bluetooth mouse

This is **IEEE 802.15.1 Bluetooth**

6. Benefits of Wi-Fi

- no cable needed
- easy internet access
- mobility
- can connect many devices
- useful in office, home, campus

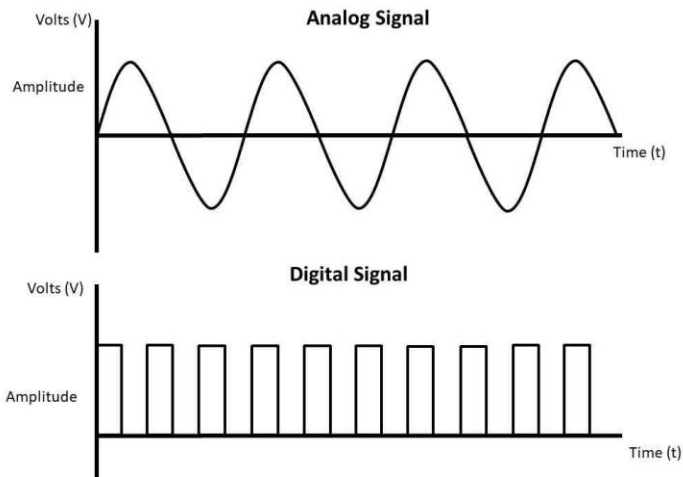
7. Benefits of Bluetooth

- simple short-range connection
- low power use
- easy pairing

- useful for personal devices
- no cable required

3.6 Signals

A **signal** is an electrical, optical, or electromagnetic form of data that carries information from one place to another.



In communication systems, data cannot travel directly. It is first converted into a signal, and then that signal is transmitted through a medium.

Simple meaning

A signal is the **representation of data** used for communication.

Example:

- human voice converted into electrical signal in telephone
- Wi-Fi data sent as radio signal
- optical fiber data sent as light signal

1. Why Signals are Needed

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Signals are needed because:

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- data must be transmitted from sender to receiver
- communication devices understand signal form
- information can travel through cable or air only in signal form

Without signals, communication is not possible.

2. Types of Signals

Signals are mainly of **two types**:

1. **Analog Signal**
2. **Digital Signal**

2.1 Analog Signal

An **analog signal** is a **continuous signal** that changes smoothly with time.

It can take infinite values within a range.

Characteristics:

- continuous in time
- continuous in amplitude
- smooth waveform
- more affected by noise

Example:

- human voice

- temperature signal
- radio broadcast
- old telephone signal

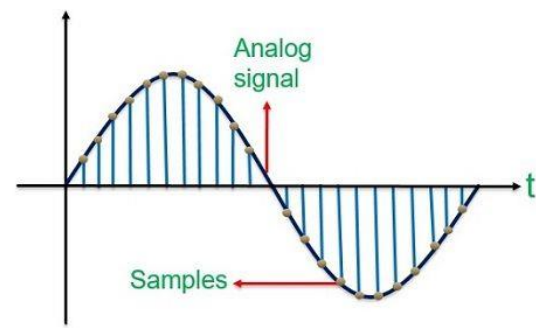
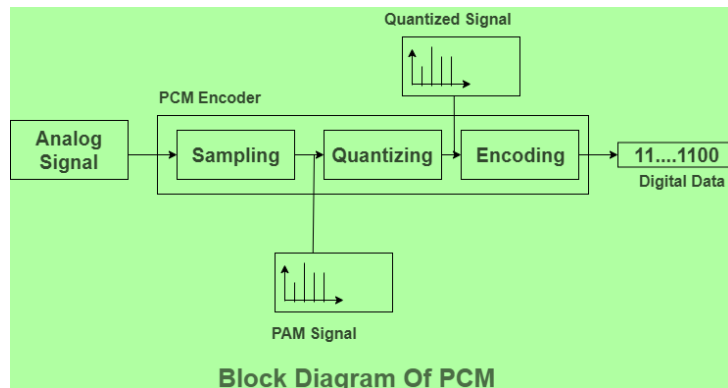
Usually shown as square pulses.

Shape:

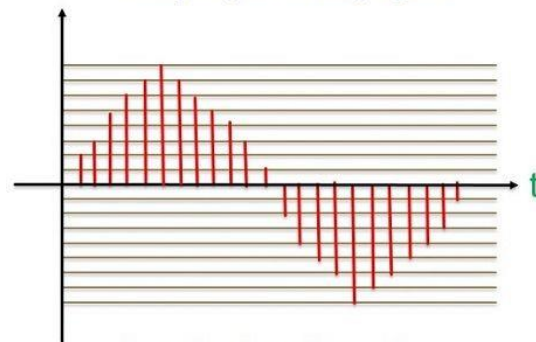
Usually shown as a smooth sine wave.

Simple example:

When a person speaks into a microphone, the sound wave produced is analog.



Sampling of analog signal



Quantization of samples

Simple example:

A computer sending bits 101101 through a network uses digital signal.

2.2 Digital Signal

A **digital signal** is a **discrete signal** that has a limited number of values, usually **0 and 1**.

Characteristics:

- discrete in nature
- represented by binary values
- less affected by noise
- easy to store and process

Example:

- computer data
- digital clock signal
- binary transmission in networks

Shape:

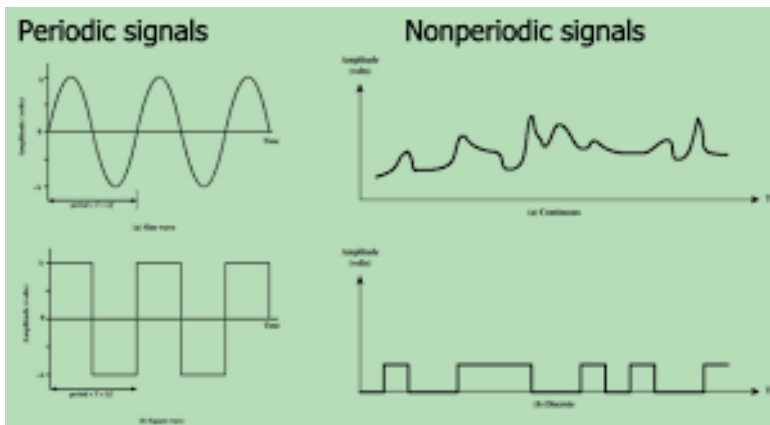
3. Difference Between Analog and Digital Signals

Basis	Analog Signal	Digital Signal
Nature	Continuous	Discrete
Values	Infinite	Limited, usually 0 and 1
Waveform	Smooth	Square pulses
Noise effect	More	Less

Basis	Analog Signal	Digital Signal
Use	Voice, radio	Computers, digital communication

4. Periodic and Non-Periodic Signals

Signals can also be classified as:



4.1 Periodic Signal

A **periodic signal** repeats itself after a fixed interval of time.

Example:

- sine wave
- clock pulse

Meaning:

The same pattern keeps repeating.

4.2 Non-Periodic Signal

A **non-periodic signal** does not repeat its pattern regularly.

Example:

- human speech

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Meaning:

The signal changes irregularly.

5. Important Terms Related to Signals

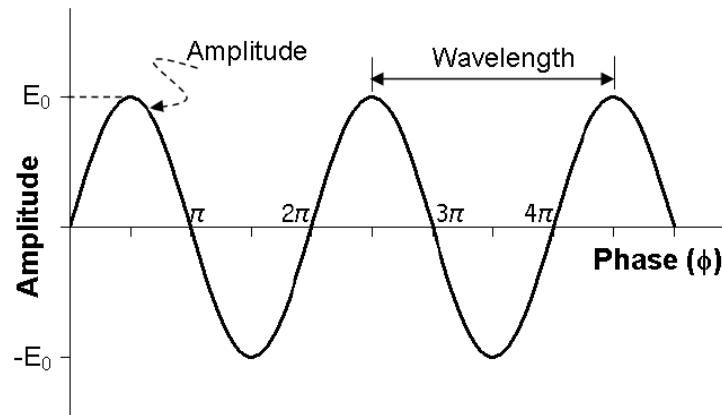
5.1 Amplitude

Amplitude is the maximum height of the signal from the reference level.

It shows the strength of the signal.

Example:

A higher voltage wave has greater amplitude.



5.2 Frequency

Frequency is the number of complete cycles of a signal in one second.

Unit:

Hertz (Hz)

Example:

If a signal completes 50 cycles in one second, its frequency is 50 Hz.

5.3 Time Period

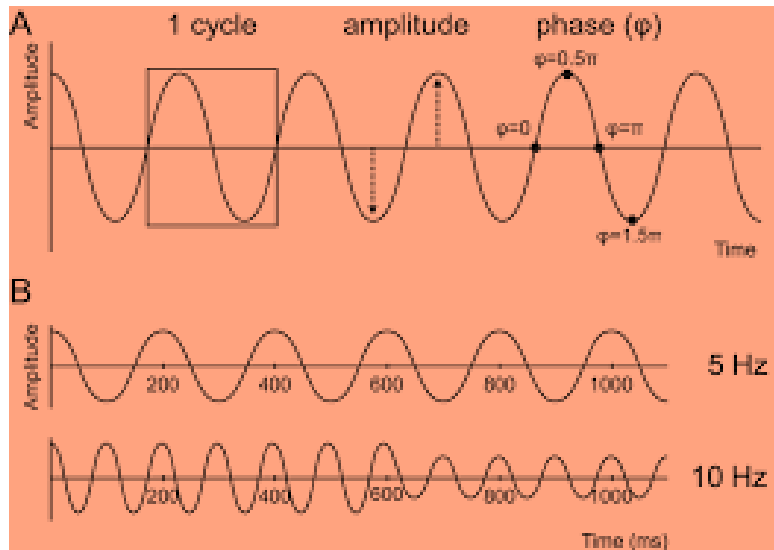
Time period is the time taken to complete one full cycle of a signal.

Formula:

$$T = 1 / f$$

Where:

- T = time period
- f = frequency



5.4 Phase

Phase tells the position of a waveform with respect to time.

It is used when comparing two signals.

6. Simple Example with Data

Suppose a signal completes **100 cycles in 1 second**.

Then,

Frequency = 100 Hz

If the signal has a maximum height of **5 V**, then:

Amplitude = 5 V

If time for one cycle is:

$$T = 1 / 100 = 0.01 \text{ second}$$

So:

- Amplitude = 5 V
- Frequency = 100 Hz
- Time period = 0.01 sec

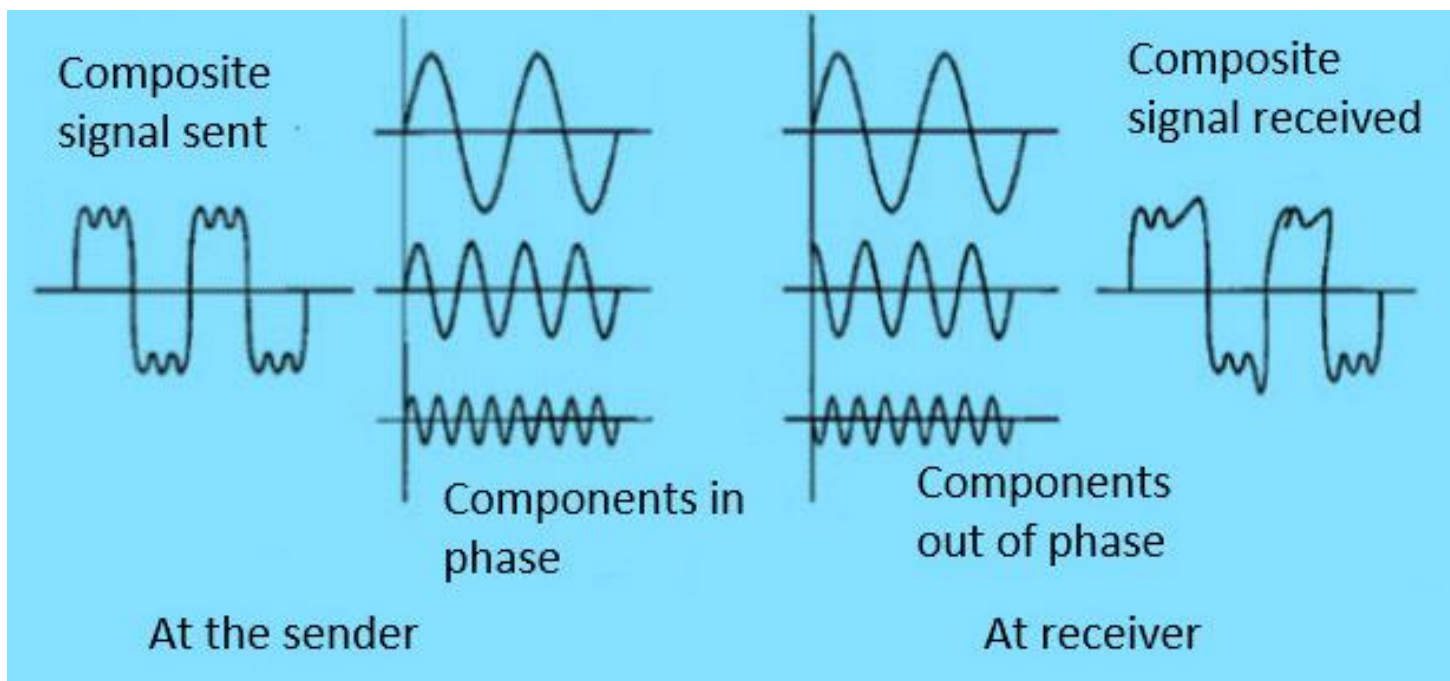
7. Real-Life Examples of Signals

Analog signal examples:

- microphone sound signal
- FM radio signal
- analog thermometer reading

Digital signal examples:

- computer data
- mobile message in binary
- digital watch output
- LAN communication



Introduction

When a signal travels from sender to receiver, it does not always remain perfect.

During transmission, the signal may become weak, changed, or mixed with unwanted signals.

These problems are called **transmission impairments**.

In simple words, transmission impairment means: **any condition that damages the quality of the signal during transmission.**

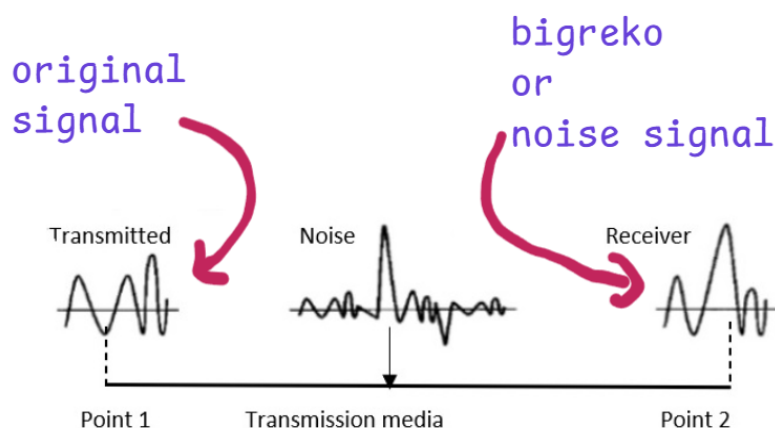
The main types of transmission impairment are:

- **Attenuation**
- **Distortion**
- **Noise**

1. Attenuation

Attenuation means the **loss of signal strength** as the signal travels through the transmission medium.

When a signal moves over a long distance, its power gradually decreases.

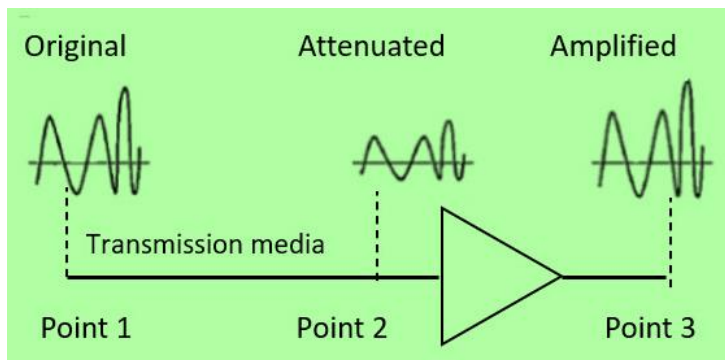


Simple meaning

The signal becomes weaker and weaker with distance.

Example:

If a signal starts with strength **10 V** at the sender and becomes **6 V** at the receiver, then attenuation has occurred.

**Causes:**

- long cable length
- resistance of medium
- energy loss in transmission

Effects:

- weak signal at receiver
- difficult to detect data correctly
- communication errors may occur

Solution:

- use **repeater**
- use **amplifier**
- use better quality medium

Daily life analogy:

Suppose Ram speaks loudly from one side of a field. As the sound travels farther, it becomes weaker. This weakening is like **attenuation**.

2. Distortion

Distortion means a **change in the shape or form of the signal** during transmission.

It happens when different frequency components of a signal travel at different speeds.

So the received signal does not look exactly like the transmitted signal.

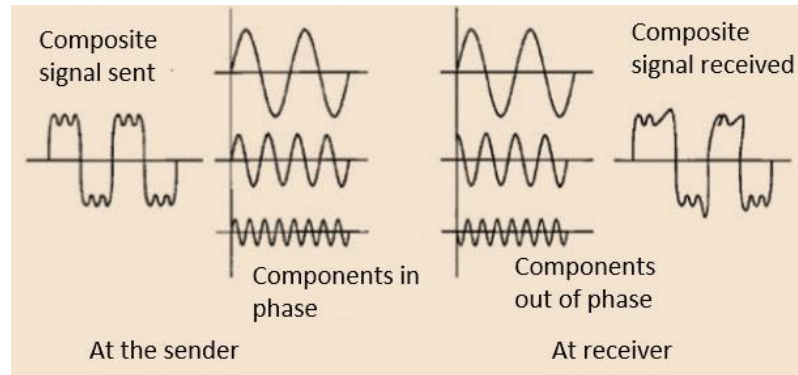
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Simple meaning

The signal reaches the receiver, but its shape becomes different.

Example:

If a signal is sent as one waveform but arrives stretched, shifted, or bent, distortion has occurred.

**Causes:**

- different propagation speeds
- medium characteristics
- bandwidth limitations

Effects:

- data may be interpreted incorrectly
- signal quality becomes poor

Daily life analogy:

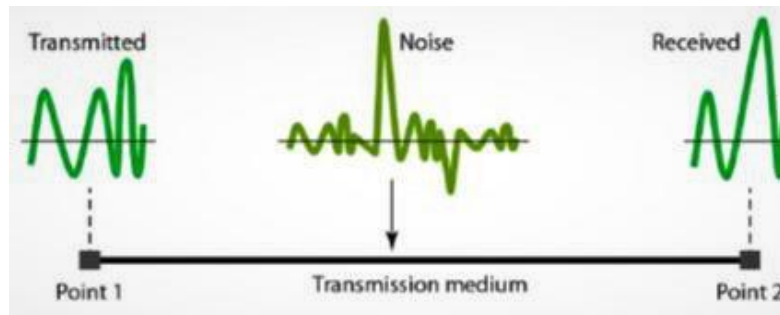
Imagine writing a straight line on paper, but after photocopying badly, it appears curved or stretched. That change in original shape is like **distortion**.

3. Noise

Noise is any **unwanted signal** that gets mixed with the original signal during transmission.

Noise disturbs communication and reduces signal quality.

Noise is extra unwanted disturbance added to the original signal.



Example:

When talking on a phone, if you hear unwanted “shhh” or cracking sound, that is noise.

Causes:

- electrical interference
- thermal effect
- atmospheric disturbances
- nearby signals
- faulty equipment

Effects:

- wrong data may be received
- signal becomes unclear
- communication quality decreases

Types of Noise

1. Thermal Noise

Caused by random movement of electrons in a conductor.

Example:

Present in almost all electronic devices.

2. Induced Noise

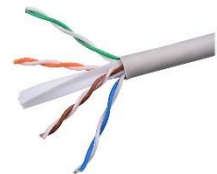
Caused by external sources such as motors, power lines, or electrical devices.

Example:

Noise from nearby electric wire affecting communication cable.

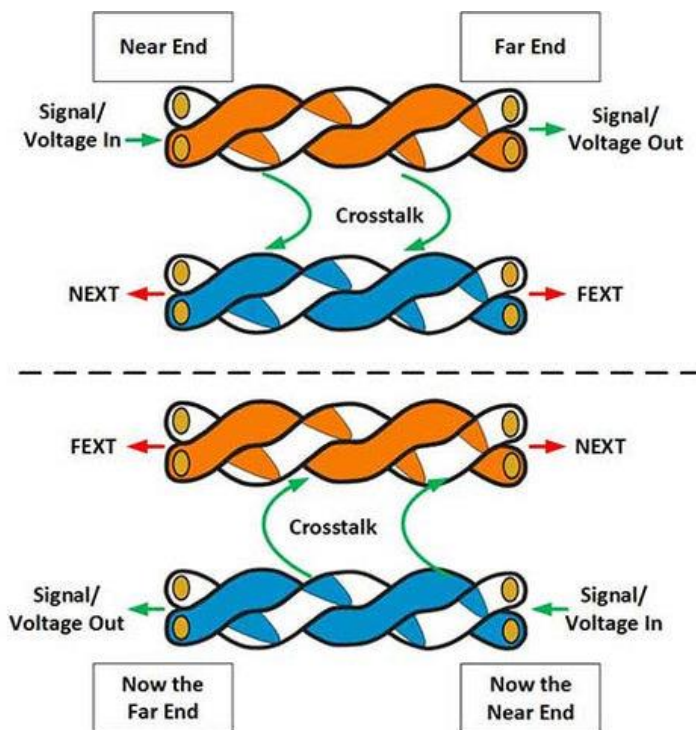
3. Crosstalk

Occurs when one signal interferes with another signal in nearby wires.



Example:

Two telephone lines affecting each other.

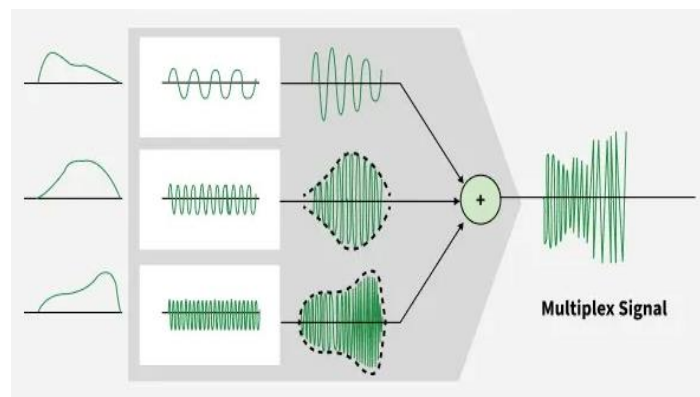


4. Impulse Noise

Sudden short-duration noise caused by lightning, switching, or faults.

Example:

A sudden spike in signal during storm.



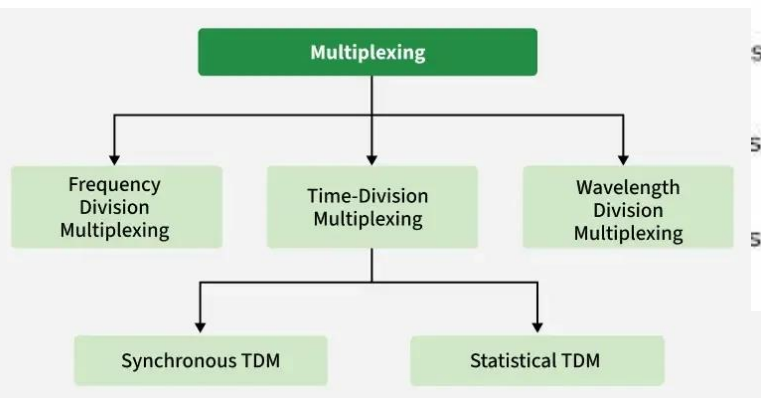
Simple flow:

Input signals → MUX → **Single channel** → DEMUX
→ Original signals

Types of Multiplexing

The main types are:

1. **TDM – Time Division Multiplexing**
2. **FDM – Frequency Division Multiplexing**
3. **WDM – Wavelength Division Multiplexing**



1. TDM – Time Division Multiplexing

In **Time Division Multiplexing (TDM)**, multiple signals share the same channel, but **each signal gets a different time slot**.

That means users do not send data at the same exact moment.

They send one after another in very small time intervals.

Simple meaning

One channel is shared by many users **by dividing time**.

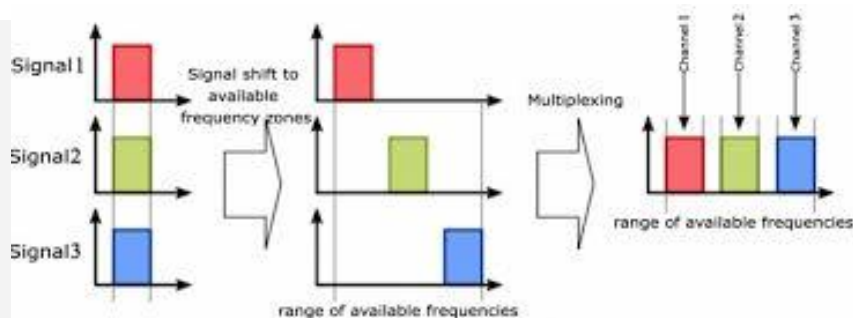
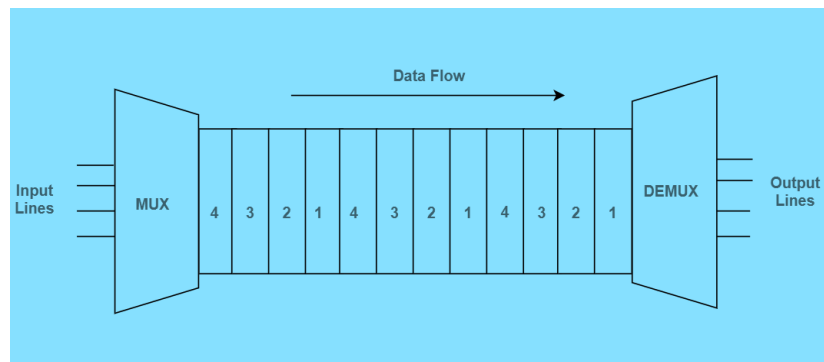
Example:

Suppose 4 users share one line:

- user 1 sends in time slot 1
- user 2 sends in time slot 2
- user 3 sends in time slot 3

- user 4 sends in time slot 4

Then the cycle repeats again.



Daily life analogy:

Imagine one water tap and 4 people filling bottles:

- first person uses the tap for 10 seconds
- second person uses it for next 10 seconds
- third person after that
- fourth person after that

Everyone uses the same tap, but at different times.
This is like **TDM**.

Uses:

- digital telephony
- computer networks
- communication systems

Advantages:

- efficient for digital signals

- no frequency separation needed
- better use of full bandwidth during time slot

Disadvantages:

- synchronization required
- if one slot is empty, time may be wasted in simple TDM

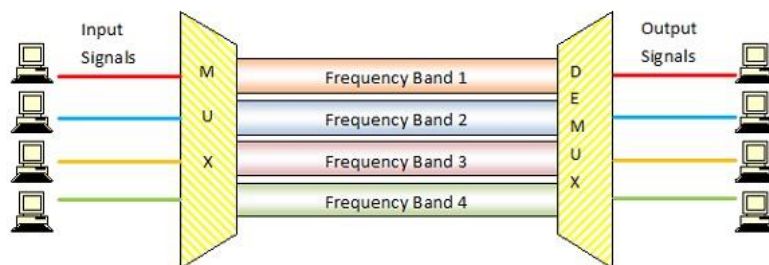
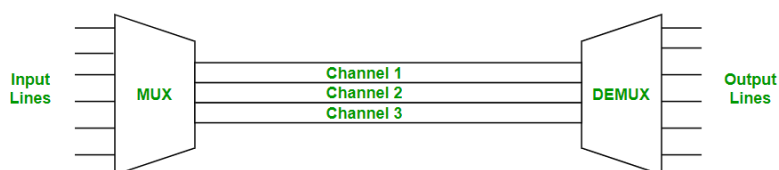
2. FDM – Frequency Division Multiplexing

In **Frequency Division Multiplexing (FDM)**, the available bandwidth is divided into different frequency bands, and each signal is assigned a separate frequency band.

All signals are transmitted **at the same time**, but on different frequencies.

Simple meaning

One channel is shared by many users **by dividing frequency**.



Example:

Different radio stations:

- Radio Nepal on one frequency

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- another FM on another frequency
- another station on another frequency

All are transmitted at the same time, but each uses a different frequency.

Daily life analogy:

Imagine a road with several lanes:

- each vehicle uses its own lane
- all vehicles move at the same time

This is like **FDM**, where each signal gets its own frequency lane.

Uses:

- radio broadcasting
- television broadcasting
- cable communication

Advantages:

- signals can transmit simultaneously
- suitable for analog transmission
- continuous transmission possible

Disadvantages:

- guard bands are needed
- bandwidth may be wasted
- interference may occur if frequencies overlap

3. WDM – Wavelength Division Multiplexing

WDM is similar to FDM, but it is used in **optical fiber communication**.

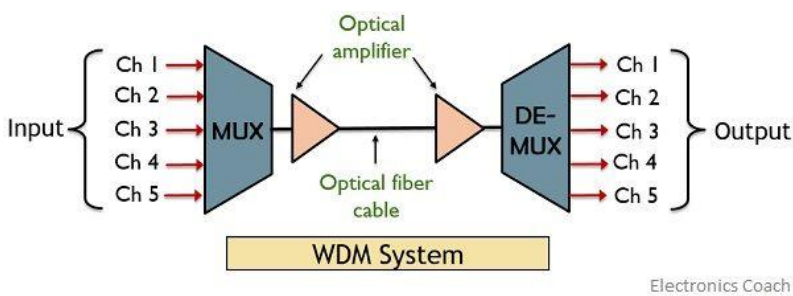
In WDM, multiple optical signals are transmitted through the same fiber using **different wavelengths of light**.

Advantages:

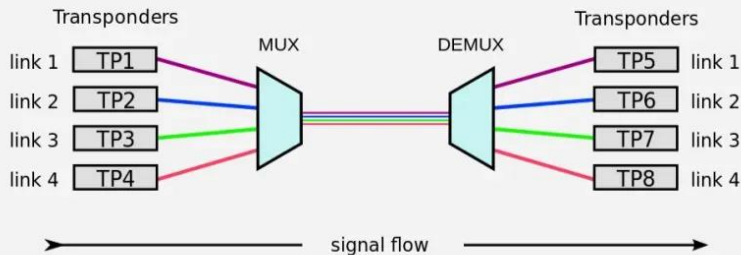
- very high capacity
- efficient use of optical fiber
- supports many channels at once

Disadvantages:

- expensive equipment
- complex design
- used mainly in fiber optic systems



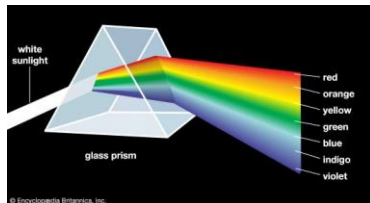
Electronics Coach

wavelength-division multiplexing (WDM)**Simple meaning**

One optical fiber carries many signals by using different light colors or wavelengths.

Example:

In fiber communication, one wavelength may carry internet data, another may carry voice, and another may carry video.

**Daily life analogy:**

Imagine white light passing through a prism and splitting into many colors. Each color can carry separate information. That idea is similar to **WDM**.

Uses:

- optical fiber networks
- internet backbone
- long-distance high-speed communication

Comparison of TDM, FDM and WDM

Basis	TDM	FDM	WDM
Full form	Time Division Multiplexing	Frequency Division Multiplexing	Wavelength Division Multiplexing
Division based on	Time	Frequency	Light wavelength
Used mainly for	Digital signals	Analog signals	Optical fiber
Transmission	One after another	Simultaneously	Simultaneously
Example	Digital phone system	Radio stations	Fiber backbone

Suppose 3 users want to send data through one channel.

In TDM:

- User 1 sends in first time slot
- User 2 sends in second time slot
- User 3 sends in third time slot

In FDM:

- User 1 gets frequency f_1
- User 2 gets frequency f_2
- User 3 gets frequency f_3

All send at the same time.

In WDM:

- User 1 gets wavelength λ_1

- User 2 gets wavelength λ_2

- User 3 gets wavelength λ_3

All send together through one optical fiber.

Small Numerical Style Example

Suppose one channel has total bandwidth of **12 kHz** and 3 signals need **4 kHz** each.

In FDM:

- Signal 1 = 4 kHz
- Signal 2 = 4 kHz
- Signal 3 = 4 kHz

Total required bandwidth:

$$4 + 4 + 4 = 12 \text{ kHz}$$

So all 3 signals can be sent together using FDM

Numerical on TDM, FDM, and WDM

1. Numerical on TDM

Question 1

A TDM system has **4 input channels**. Each channel is given **1 time slot** in one frame.

If each time slot carries **8 bits**, find:

1. Total number of bits in one frame
2. If 1000 frames are sent per second, find the total bit rate

Solution

Given:

- Number of channels = 4

- Bits per slot = 8

- Frames per second = 1000

1. Bits in one frame

Each frame contains one slot from each channel.

So,

Bits per frame = Number of channels $\times$ Bits per slot

$$\text{Bits per frame} = 4 \times 8 = 32 \text{ bits}$$

2. Bit rate

Bit rate = Bits per frame $\times$ Frames per second

$$\text{Bit rate} = 32 \times 1000 = 32000 \text{ bps}$$

Answer

- Bits in one frame = **32 bits**
- Total bit rate = **32000 bps** or **32 kbps**

Question 2

In a TDM system, **8 devices** share one channel.
Each device sends **10 bits** in one frame.
Find the number of bits in one frame.

Solution

Given:

- Number of devices = 8
- Bits per device per frame = 10

Bits per frame = $8 \times 10 = 80$ bits

Answer

Bits in one frame = 80 bits

2. Numerical on FDM

Question 1

An FDM system is designed to combine **3 voice channels**, each requiring **4 kHz** bandwidth.
Find the total required bandwidth if guard bands are ignored.

Solution

Given:

- Number of channels = 3
- Bandwidth per channel = 4 kHz

Total bandwidth = Number of channels $\times$ Bandwidth per channel

Total bandwidth = $3 \times 4 = 12$ kHz

Answer

Total required bandwidth = 12 kHz

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Question 2

A communication system uses FDM to combine **5 channels**.

Each channel requires **6 kHz** bandwidth.

There is a **1 kHz guard band** between adjacent channels.

Find the total bandwidth required.

Solution

Given:

- Number of channels = 5
- Bandwidth per channel = 6 kHz
- Guard band = 1 kHz

Step 1: Total channel bandwidth

$$5 \times 6 = 30 \text{ kHz}$$

Step 2: Number of guard bands

Between 5 channels, guard bands = 4

Step 3: Total guard band bandwidth

$$4 \times 1 = 4 \text{ kHz}$$

Step 4: Total bandwidth

$$\text{Total bandwidth} = 30 + 4 = 34 \text{ kHz}$$

Answer

Total required bandwidth = 34 kHz

3. Numerical on WDM

Question 1

A WDM system uses **4 optical channels** on one fiber.
If each channel carries **2.5 Gbps**, find the total data carrying capacity of the fiber.

Solution

Given:

- Number of channels = 4

- Capacity per channel = 2.5 Gbps

Total capacity = Number of channels × Capacity per channel

$$\text{Total capacity} = 4 \times 2.5 = 10 \text{ Gbps}$$

Answer

Total fiber capacity = 10 Gbps

Question 2

A WDM optical fiber carries **8 wavelengths**, and each wavelength carries **10 Gbps**.

Find the total transmission capacity.

Solution

Given:

- Number of wavelengths = 8
- Capacity per wavelength = 10 Gbps

$$\text{Total capacity} = 8 \times 10 = 80 \text{ Gbps}$$

Answer

Total transmission capacity = 80 Gbps

Mixed Short Practice Set

TDM Practice

A TDM frame has **6 channels**, each contributing **7 bits**.
Find bits per frame.

Solution

$$\text{Bits per frame} = 6 \times 7 = 42 \text{ bits}$$

Answer

42 bits

FDM Practice

An FDM system combines **4 channels**, each of **5 kHz**.
Find total bandwidth without guard bands.

Solution

$$\text{Total bandwidth} = 4 \times 5 = 20 \text{ kHz}$$

Answer

20 kHz

WDM Practice

A WDM system has **16 wavelengths**, each carrying **5 Gbps**.

Find total capacity.

Solution

$$\text{Total capacity} = 16 \times 5 = 80 \text{ Gbps}$$

Answer

80 Gbps

Short Formulas

TDM

Bits per frame = Number of channels × Bits per slot

Bit rate = Bits per frame × Frames per second

FDM

Total bandwidth = Sum of channel bandwidths + Guard bands

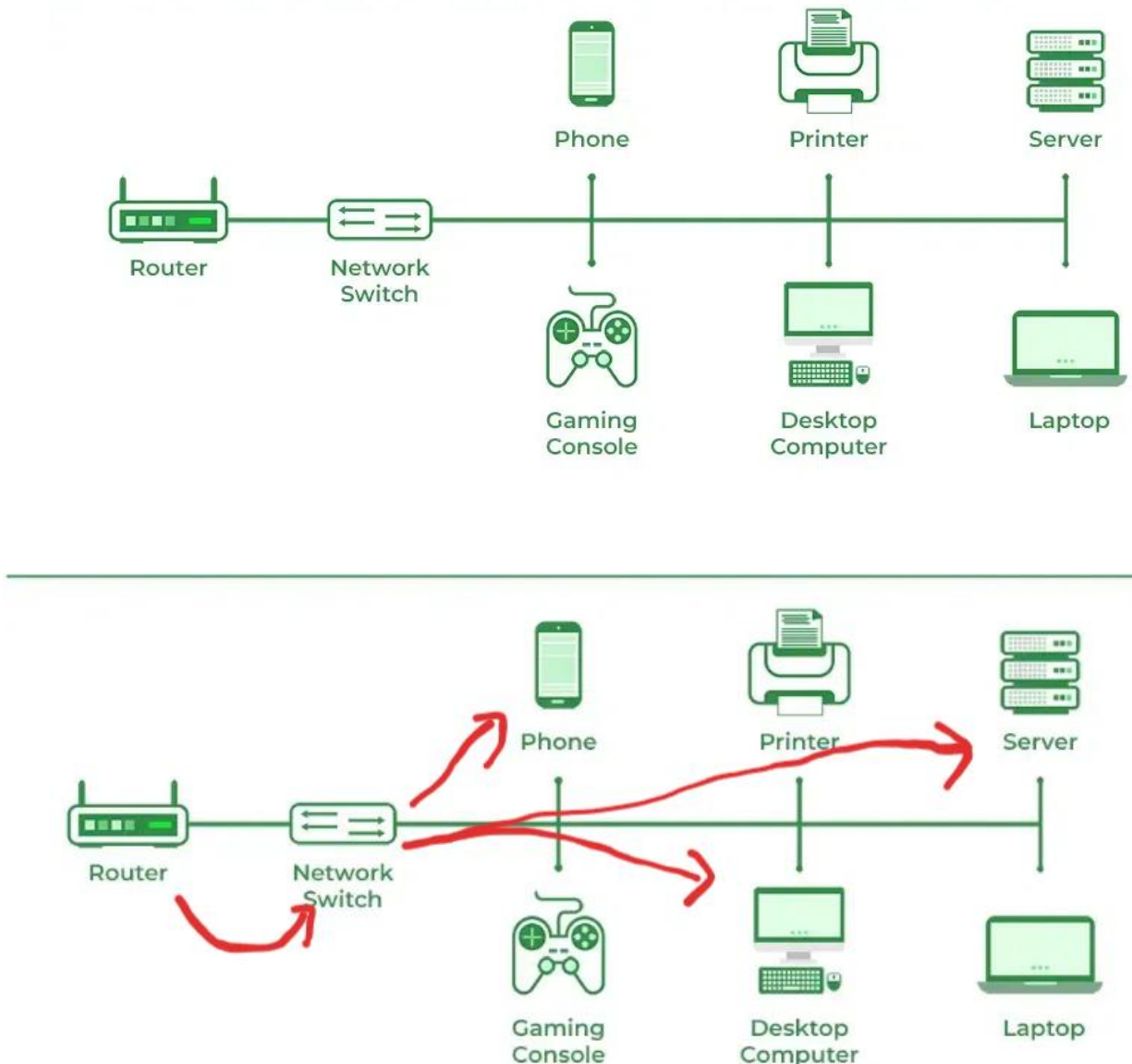
If equal bandwidth:

Total bandwidth = n × channel bandwidth + guard bands

3.9 Switching: Circuit Switching, Message Switching, Packet Switching, Virtual Circuit Switching

Network Switching is a specific type of switching used in computer networks. It is about transferring data packets from one device (node) to another across a network.

How Does a Network Switch Works?

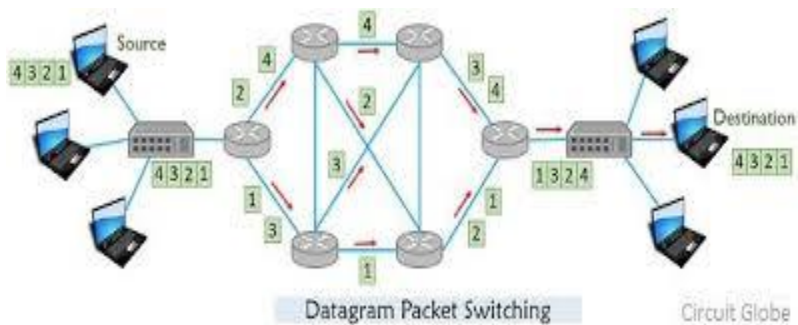


Switching is the process of transferring data from a **source to a destination** through a network.

In a communication network, many devices are connected. So the network must decide:

- how data should travel
- which path should be used
- how sender and receiver should communicate

This process is called **switching**.



Simple meaning

Switching means **choosing and using a path** to send data from one device to another.

Example:

When you send a message from your mobile phone to another person, the network chooses a path to deliver that message.

Why Switching is Needed

Switching is needed because:

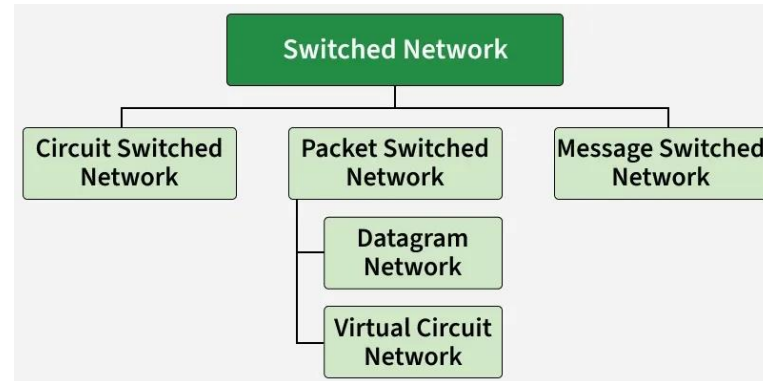
- every user cannot have a separate direct line to every other user
- network resources must be shared
- data must reach the correct destination
- communication must be efficient and organized

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Types of Switching

The main types of switching are:

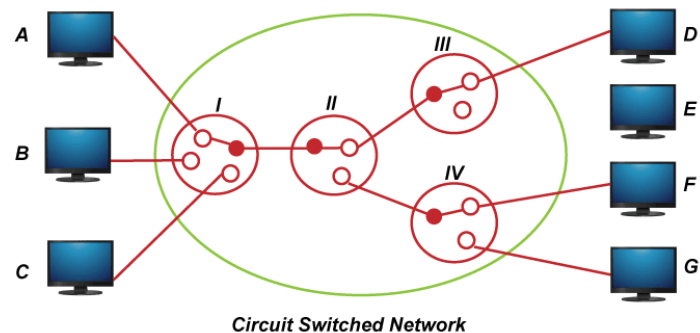
1. **Circuit Switching**
2. **Message Switching**
3. **Packet Switching**
4. **Virtual Circuit Switching**



1. Circuit Switching

In **circuit switching**, a **dedicated communication path** is established between sender and receiver before data transfer begins.

The path remains reserved for the whole communication session.



Simple meaning

First a path is created, then communication happens on that fixed path.

Traditional telephone system

When two people make a phone call, a dedicated path is established between them and remains active until the call ends.

Phases of Circuit Switching

There are usually three phases:

1. Setup Phase

The path is established.

2. Data Transfer Phase

Actual communication takes place.

3. Disconnect Phase

The path is released after communication ends.

Advantages of Circuit Switching

- dedicated path available
- continuous transmission
- good for voice communication
- no need to decide path for each data unit

Disadvantages of Circuit Switching

- channel remains reserved even when no data is being sent
- bandwidth may be wasted
- setup time is required
- not efficient for bursty data communication

Imagine reserving an entire road lane only for Ram's car from Butwal to Kathmandu.

Even if Ram stops for tea, the lane stays reserved for him.

That is like **circuit switching**.

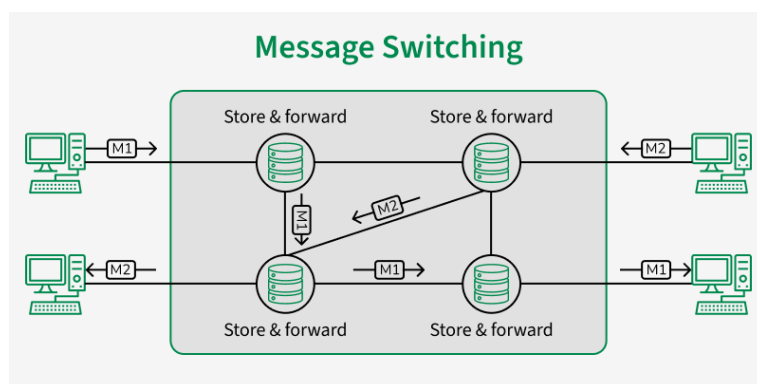
2. Message Switching

In **message switching**, the **entire message is treated as one unit** and sent from one node to another.

Each intermediate node:

- receives the whole message
- stores it
- then forwards it to the next node

This is called **store-and-forward switching**.

**Simple meaning**

The complete message is stored at each stop before being sent forward.

Example:

Telegram-like communication system or older communication methods.

Advantages of Message Switching

- no dedicated path needed
- efficient channel use

- messages can be routed dynamically

Disadvantages of Message Switching

- high delay
- not suitable for real-time communication
- large storage needed at intermediate nodes

Daily Life Analogy

Suppose Sita sends a full parcel through a courier office. The parcel reaches one office, stays there for some time, then goes to the next office, and finally reaches the destination.

That is like **message switching**.

3. Packet Switching

In **packet switching**, the message is divided into **small units called packets**.

Each packet is sent separately through the network. Packets may take:

- the same path
- or different paths

At the destination, all packets are reassembled to form the original message.

Simple meaning

Big message is broken into smaller packets and sent individually.

Example:

Internet communication

When you browse a website or send an email, the data is broken into packets.

Advantages of Packet Switching

- efficient use of bandwidth
- suitable for data communication
- no dedicated path needed
- supports many users at the same time
- reliable and flexible

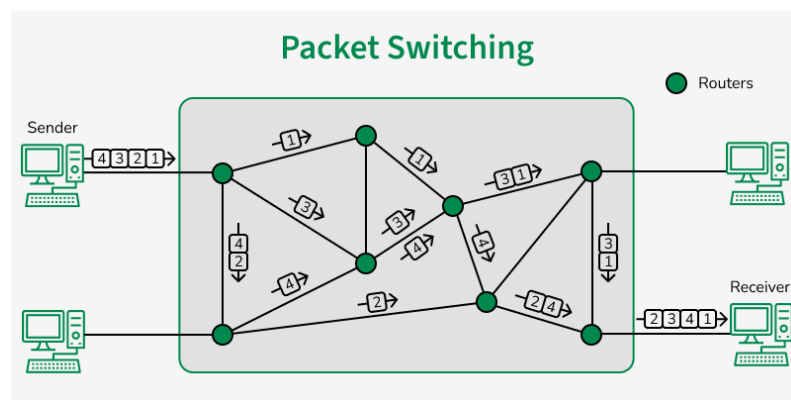
Disadvantages of Packet Switching

- packets may arrive out of order
- delay may vary
- packets may be lost and need retransmission

Daily Life Analogy

Suppose Ram wants to send a book to Hari, but instead of sending it as one big parcel, he tears it into chapters, packs each chapter separately, and sends them one by one. Hari later combines them in correct order.

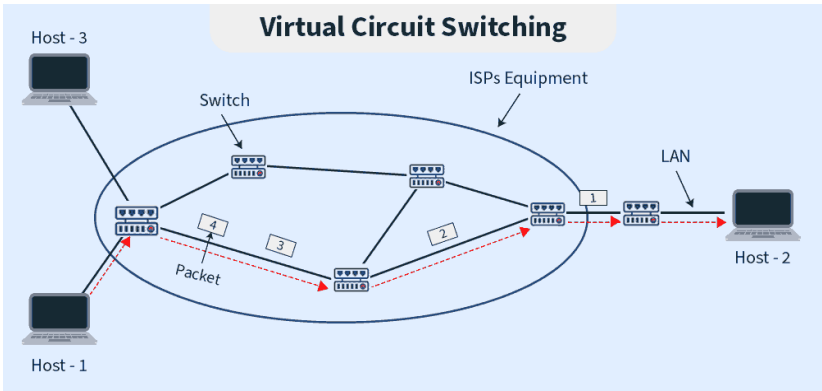
That is like **packet switching**.



4. Virtual Circuit Switching

Virtual circuit switching is a type of packet switching in which a **logical path** is established before packets are sent.

All packets follow the same logical route, but the network does not reserve a dedicated physical path like circuit switching.



Simple meaning

A route is decided first, and then all packets follow that same route.

Example:

Frame relay, ATM, and some WAN technologies

Advantages of Virtual Circuit Switching

- packets arrive in order
- easier management than pure packet switching
- better than message switching for many applications

Disadvantages of Virtual Circuit Switching

- setup phase required
- if path fails, communication is affected
- less flexible than pure packet switching

Daily Life Analogy

Imagine a courier company deciding one fixed route for all of Ram's parcels before delivery starts. Each small parcel follows that same decided route.

That is like **virtual circuit switching**.

Comparison of Switching Techniques

Switching Type	Path	Data Unit	Delay	Best Use
Circuit Switching	Dedicated physical path	Continuous stream	Low after setup	Telephone/voice
Message Switching	No dedicated path	Entire message	High	Non-real-time
Packet Switching	No dedicated path	Packets	Variable	Internet/data
Virtual Circuit Switching	Logical fixed path	Packets	Moderate	WAN/data networks

Numerical 1: Circuit Switching Case

A college in Butwal has connected its administration office with the examination section through a circuit-switched communication system. During an important result-processing day, Ram from the administration office needs to talk continuously with Sita in the examination section for **12 minutes**. The communication line provides a dedicated data path of **64 kbps** for the whole session. Once the circuit is established, the entire bandwidth remains reserved until the conversation ends, even if the users remain silent for some time. The network administrator wants to calculate how much total data-carrying capacity was reserved by the network during the full call duration. Assume the reserved channel remains active for the complete 12 minutes.

Given:

- Bandwidth = **64 kbps**
- Time = **12 minutes**

Solution:

Convert time into seconds:

$$12 \text{ minutes} = 12 \times 60 = 720 \text{ seconds}$$

Now,

$$\text{Total data capacity reserved} = \text{Bandwidth} \times \text{Time}$$

$$= 64 \text{ kbps} \times 720 \text{ s}$$

$$= 64000 \text{ bps} \times 720$$

$$= 46,080,000 \text{ bits}$$

Convert into kilobits:

$$46,080,000 / 1000 = 46,080 \text{ kb}$$

Convert into megabits:

$$46,080,000 / 1,000,000 = 46.08 \text{ Mb}$$

Numerical 2: Message Switching Case

A district office in Nepal wants to send a large digital notice file from one branch to another using a **message switching** network. The full message size is **900 KB**. The message must pass through **3 intermediate switching nodes** before reaching the final destination. In message switching, the **entire message is first received, stored, and then forwarded** at each node. Assume that the transmission speed of every link is **300 KB per second**, and ignore propagation delay. The office wants to know the total time needed for the whole message to reach the destination. Since the message must pass through 3 intermediate nodes, it will travel over **4 links total**: source to node 1, node 1 to node 2, node 2 to node 3, and node 3 to destination.

Given:

- Message size = **900 KB**
- Transmission speed = **300 KB/s**
- Number of links = **4**

Solution:

Time to transmit the whole message over one link:

$$\begin{aligned} \text{Time} &= \text{Message size} / \text{Transmission speed} \\ &= 900 \text{ KB} / 300 \text{ KB/s} = 3 \text{ seconds} \end{aligned}$$

Since message switching uses store-and-forward, the full message must be sent completely over each link.

Total time:

$$\begin{aligned} \text{Total time} &= 3 \text{ seconds} \times 4 \text{ links} \\ &= 12 \text{ seconds} \end{aligned}$$

Answer:

The total data-carrying capacity reserved during the 12-minute circuit-switched session is:

46,080,000 bits

or

46,080 kb

or

46.08 Mb

Conclusion:

In circuit switching, the full channel remains reserved for the complete duration, so even if no actual data is sent at some moments, the capacity is still occupied.

The total time needed to send the full message to the destination is: **12 seconds**

Conclusion:

In message switching, the whole message is stored at each intermediate node before forwarding, so delay becomes high. That is why message switching is not suitable for real-time communication.

Numerical 3: Packet Switching Case

Hari is sending a project report from his college computer to the central server using a **packet switching** network. The total report size is **1200 KB**. The network divides this report into packets of **200 KB each**. The transmission speed of the link is **400 KB per second**. The path contains **2 intermediate routers**, which means the packets pass through **3 links total** from source to destination. Assume **store-and-forward packet switching**, and ignore propagation delay. Hari wants to know two things:

1. how many packets will be created, and
2. how much time will be required for the **first packet** to reach the destination and for the **entire file** to be delivered, assuming pipelining is used.

Given:

- Total file size = **1200 KB**
- Packet size = **200 KB**
- Transmission speed = **400 KB/s**

Question 1: Circuit Switching

A college in Butwal has connected its administration office with the examination section through a **circuit-switched communication system**. During result preparation, Ram from the administration office needs to communicate continuously with Sita in the examination section for **15 minutes**. The communication channel provides a dedicated bandwidth of **64 kbps** for the entire session. Once the circuit is established, the full bandwidth remains reserved until the communication ends, even if no actual data is being sent during some moments. Calculate the **total data-carrying capacity reserved** by the network during the full communication period.

Question 2: Message Switching

A government office wants to send a digital notice file of size **1200 KB** from one branch office to another using a **message switching network**. The message must pass through **2 intermediate switching nodes** before reaching the final destination. The transmission speed of each link is **300 KB per second**. In message switching, the complete message is received, stored, and then forwarded at each intermediate node. Ignoring propagation delay, calculate the **total time required** for the complete message to reach the destination.

- Number of links = 3

Solution:

Step 1: Number of packets

$$\text{Number of packets} = \text{Total file size} / \text{Packet size} = 1200 / 200 = 6 \text{ packets}$$

Step 2: Time to transmit one packet on one link

$$\text{Time} = \text{Packet size} / \text{Transmission speed} = 200 / 400 = 0.5 \text{ second}$$

Step 3: Time for first packet to reach destination

The first packet must cross 3 links:

$$\text{First packet arrival time} = 3 \times 0.5 = 1.5 \text{ seconds}$$

Step 4: Time for all packets to arrive

After the first packet arrives, the remaining packets follow in pipelined form.

For n packets over m links:

$$\text{Total time} = (m + n - 1) \times \text{transmission time per packet}$$

Here:

- $m = 3$ $n = 6$
- transmission time per packet = 0.5 s

So,

$$\text{Total time} = (3 + 6 - 1) \times 0.5 = 8 \times 0.5 = 4 \text{ seconds}$$

Answer:

- Number of packets = 6
- First packet reaches destination in 1.5 seconds

Question 3: Packet Switching

Hari is sending a project report of size **1000 KB** from his college computer to the central server using a **packet switching network**. The report is divided into packets of **250 KB each**. The transmission speed of each link is **500 KB per second**, and the path contains **2 intermediate routers**, making a total of **3 links** from source to destination. Assume **store-and-forward packet switching** and ignore propagation delay. Calculate:

1. the **number of packets created**, and
2. the **total time required** for the entire report to reach the destination using pipelining.

- Entire file reaches destination in **4 seconds**

Conclusion:

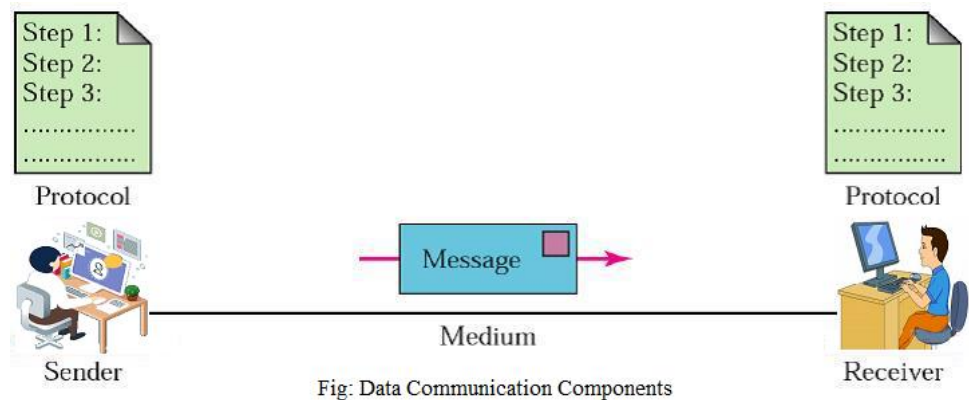
Packet switching is faster than message switching for large data because packets move in pipeline form, allowing efficient use of the network.

3.10 Network Performance: Bandwidth, Throughput, Latency, Bandwidth-Delay Product, Jitter

Network performance means how well a network works while sending and receiving data.

A good network should:

- send data fast
- deliver data correctly
- have less delay
- have stable communication



To measure network performance, some important terms are used:

- | | | | | |
|--------------------|-------------------|----------------|--------------------------------|---------------|
| • Bandwidth | Throughput | Latency | Bandwidth-Delay Product | Jitter |
|--------------------|-------------------|----------------|--------------------------------|---------------|

1. Bandwidth

Bandwidth is the **maximum capacity** of a communication channel to carry data.

In simple words, it tells the **maximum amount of data** that can be sent through a network in one second.

Unit:

- bps = bits per second
- kbps, Mbps, Gbps

Example:

If a network link supports **100 Mbps**, then its maximum capacity is 100 megabits per second.

Simple analogy:

Think of bandwidth like the **width of a water pipe**.

- wider pipe = more water can flow
- larger bandwidth = more data can flow

Important point:

Bandwidth shows the **maximum possible rate**, not the actual rate.

2. Throughput

Throughput is the **actual amount of data successfully transferred** through the network in a given time.

It is the real performance of the network.

Unit:

- bps
- kbps
- Mbps

Example:

A network may have bandwidth of **100 Mbps**, but because of traffic, errors, and overhead, actual data transfer may be only **75 Mbps**.

Then:

- Bandwidth = 100 Mbps
- Throughput = 75 Mbps

Simple analogy:

If a road can support 100 cars per minute, but because of traffic only 70 cars actually pass, then:

- road capacity = bandwidth
- actual cars passed = throughput

Important point:

Throughput is usually less than bandwidth.

3. Latency

Latency is the **total time taken by data to travel from source to destination**.

It is also called **delay**.

Unit:

- **milliseconds (ms)**
- **seconds**

Example:

If a packet takes **20 ms** to travel from one computer to another, then the latency is **20 ms**.

Causes of latency:

- propagation delay
- transmission delay
- processing delay
- queuing delay

Simple analogy:

Suppose you send a letter today and it reaches tomorrow.

The time taken is like **latency**.

Important point:

Lower latency means better performance, especially for:

- gaming
 - video calls
 - online classes
 - live streaming
-

4. Bandwidth-Delay Product

The **Bandwidth-Delay Product** is the amount of data that can be “in transit” in the network at one time.

It is calculated by:

Formula:

Bandwidth-Delay Product = Bandwidth × Delay

Meaning:

It shows how much data the network can hold while data is moving from sender to receiver.

Example:

If:

- Bandwidth = **1 Mbps**
- Delay = **0.1 second**

Then:

Bandwidth-Delay Product = 1,000,000 × 0.1 = 100,000 bits

This means **100,000 bits** can be present in the network at one time.

Simple analogy:

Think of a pipe carrying water.

- bandwidth = how much water per second can flow
- delay = how long water takes to reach the end
- bandwidth-delay product = how much water is inside the pipe

5. Jitter

Jitter is the **variation in packet arrival time**.

In simple words, if packets do not arrive at equal intervals, jitter occurs.

Example:

Suppose 3 packets arrive as:

- Packet 1 = 10 ms
- Packet 2 = 20 ms
- Packet 3 = 12 ms

The arrival time is not consistent, so jitter is present.

Effects of jitter:

- poor voice quality
- broken video
- interruption in real-time communication

Used in:

- VoIP
- video conferencing
- online gaming
- streaming

Simple analogy:

Suppose a bus is expected every 10 minutes, but comes after 8 minutes, then 14 minutes, then 9 minutes. This variation is like **jitter**.

Comparison of Terms

Term	Meaning	Unit
Bandwidth	Maximum capacity of network	bps
Throughput	Actual data transfer rate	bps
Latency	Time taken for data to travel	ms / sec

Term	Meaning	Unit
Bandwidth-Delay Product	Data present in network during transmission	bits
Jitter	Variation in packet arrival time	ms

Simple Example with Data

Suppose a network has:

- Bandwidth = 50 Mbps
- Throughput = 35 Mbps
- Latency = 30 ms

This means:

- maximum capacity is 50 Mbps
- actual transferred data is 35 Mbps
- one packet takes 30 ms to reach destination

If packet arrival times are:

- 30 ms
- 40 ms
- 28 ms

Then there is **jitter**.

Small Numerical Examples

Numerical 1: Throughput

A network link has bandwidth 100 Mbps, but actual transfer is only 80 Mbps.

Answer:

- Bandwidth = 100 Mbps
- Throughput = 80 Mbps

Numerical 2: Latency

A packet is sent at 2:00:00 and received at 2:00:00.050

Answer:

Latency = 50 ms

Numerical 3: Bandwidth-Delay Product

Bandwidth = 2 Mbps

Delay = 0.2 sec

Bandwidth-Delay Product = $2,000,000 \times 0.2 = 400,000$ bits

Answer:

400,000 bits

Why These Terms Matter

These network performance terms are important because they help us:

- evaluate network quality
- compare networks
- improve speed and reliability
- troubleshoot problems
- design better communication systems

